

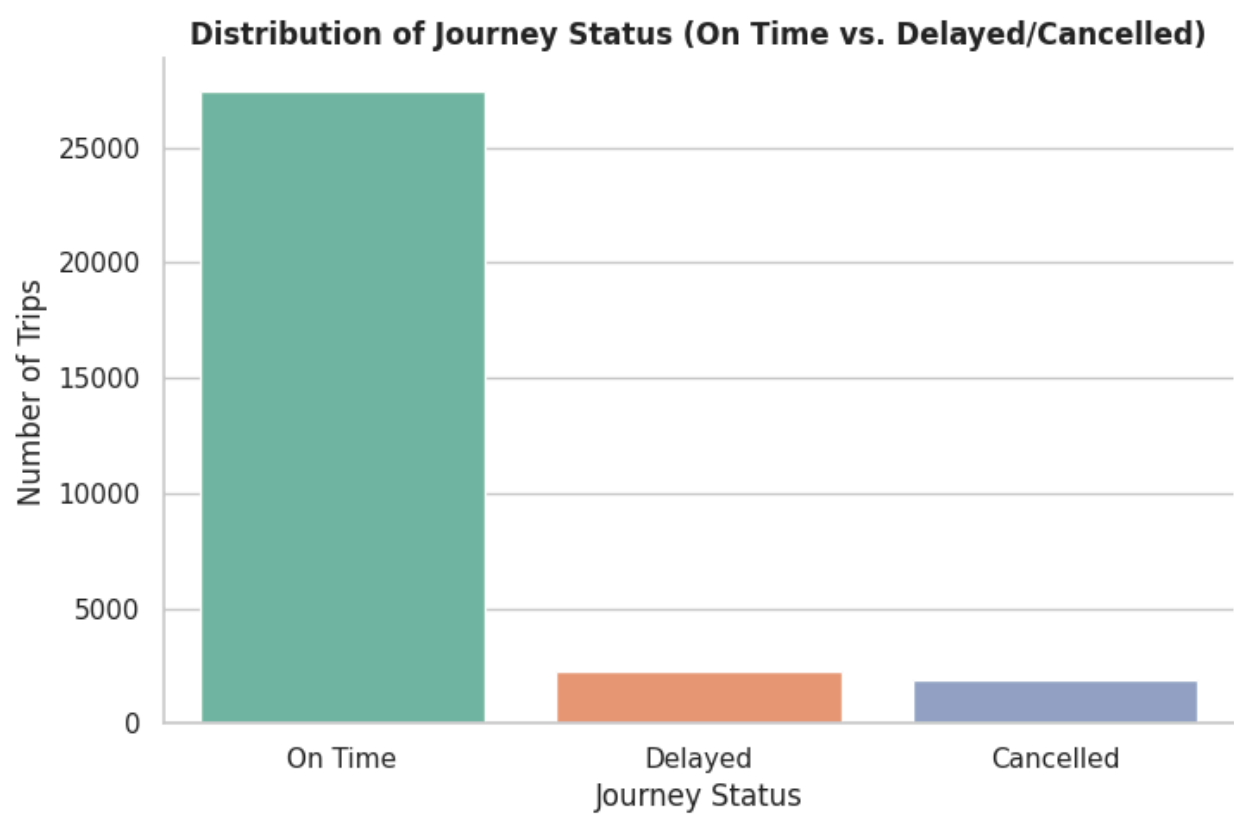


Technical Report

1 Univariate Analysis

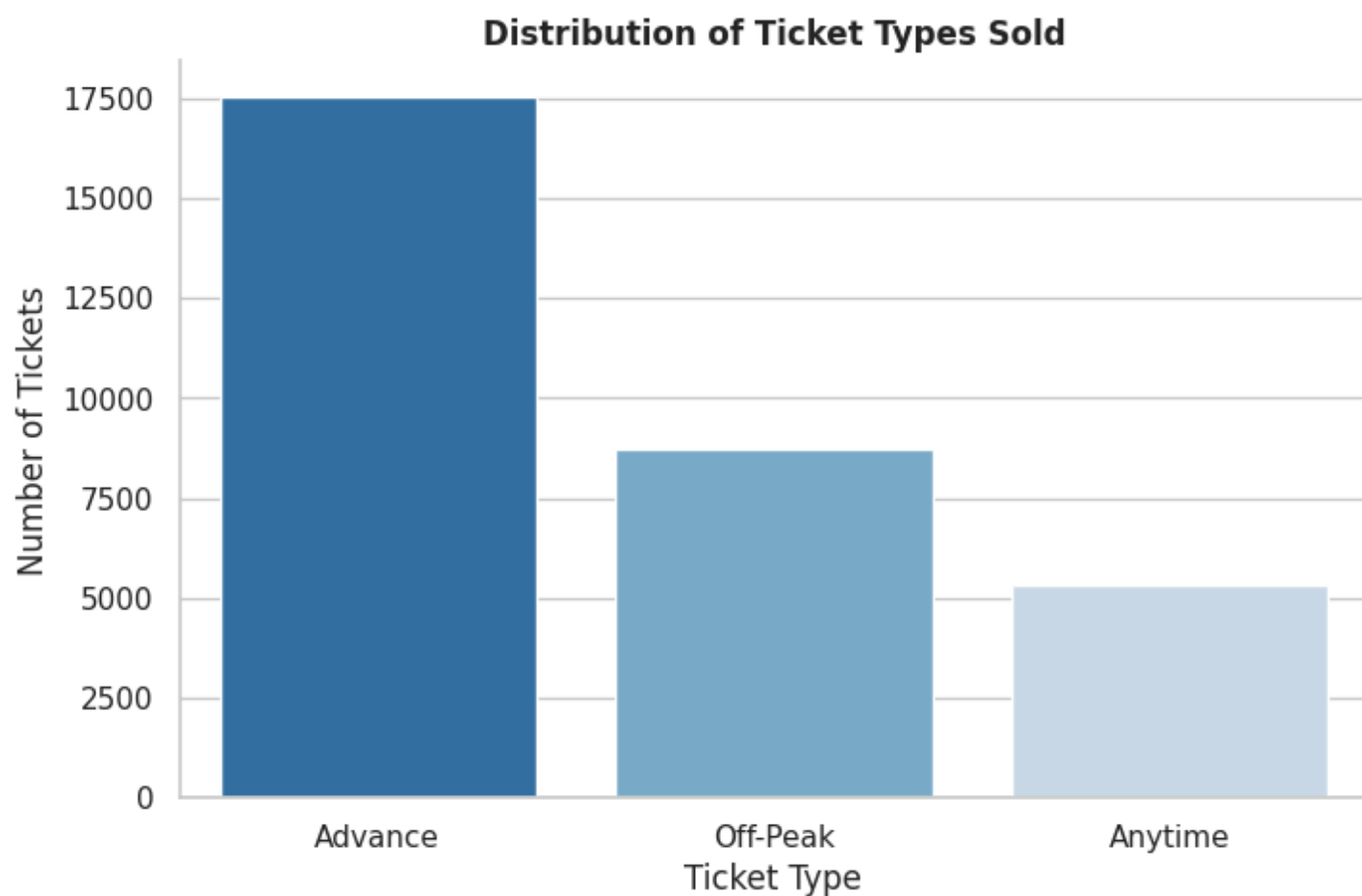
In this initial section, we explore the individual distributions of our key variables to understand the baseline operational and financial environment before moving into complex correlations.

Q1: What is the overall distribution of operational performance (Journey Status) across the network?

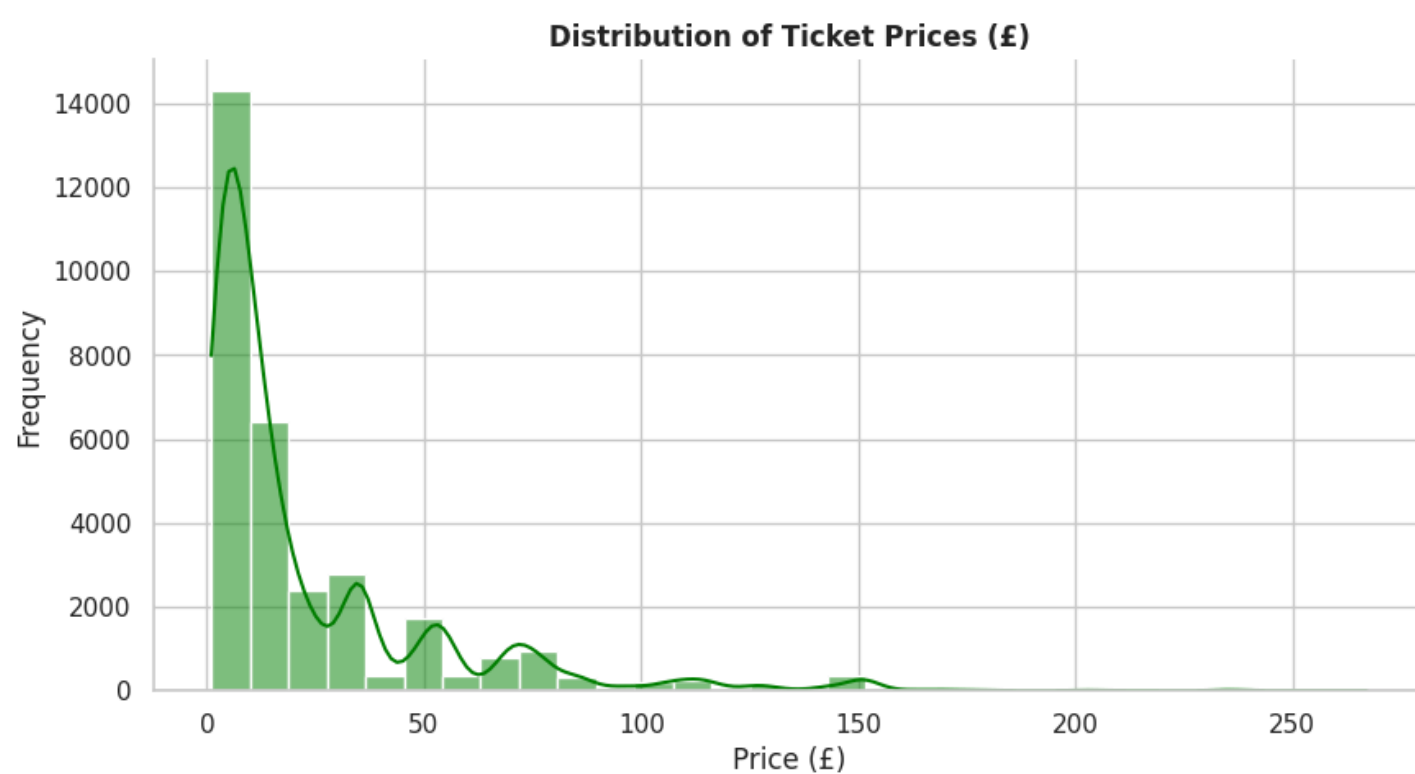


Finding: The univariate analysis of Journey Status indicates that the vast majority of our network's trips arrive **On Time**. However, there is a visually significant volume of **Delayed** trips that require deeper operational investigation in the subsequent sections to determine the root causes.

Q2: What is the most popular Ticket Type purchased by our passengers?



Q3: How are ticket prices distributed, and what is the most common price range?



Finding: The distribution of ticket prices is heavily right-skewed. The bulk of our tickets are sold at lower price points (under £50), with a very long tail of premium, high-priced tickets reaching upwards of £200. This confirms that our core revenue is driven by high-volume, low-margin transactions rather than premium sales.

2 Bivariant Analysis

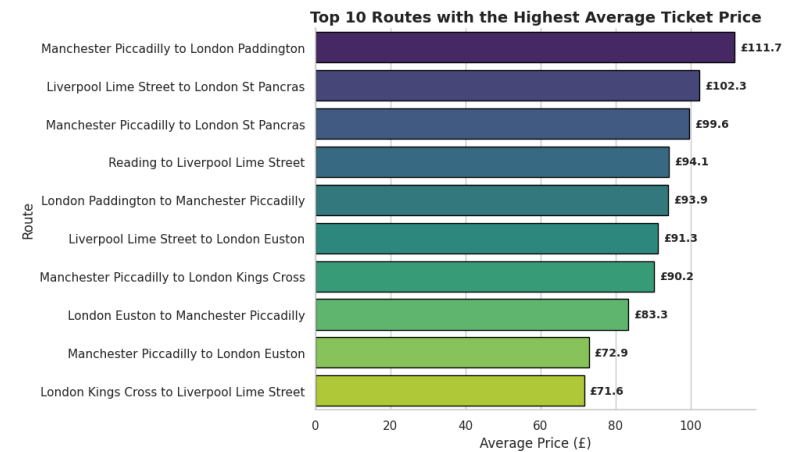
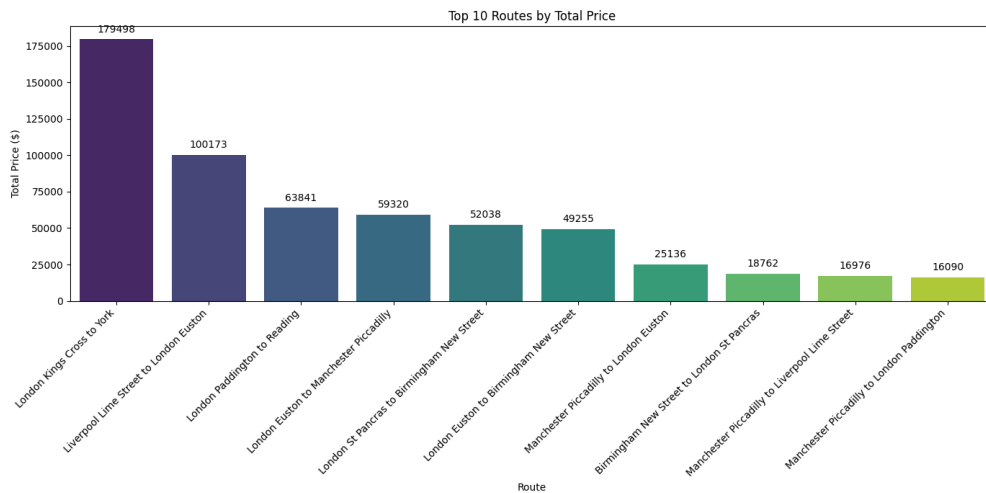
In this section, we focus on Bivariate Analysis, which primarily involves examining the relationship between two distinct variables simultaneously. The main objective is to uncover underlying patterns and identify cause-and-effect relationships within the data, such as the impact of operational delay reasons on refund requests or how the ticket class influences total revenue. We have formulated 16 core Business Questions

Pillar 1

In this section, we analyze the core drivers of the company's profitability. By evaluating revenue across different routes, ticket classes, and purchasing timelines (lead time), we aim to identify opportunities to optimize yield management and reduce margin leakage caused by excessive discounting

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Q1: Which route generates the highest net revenue and highest average ticket prices for the company ?

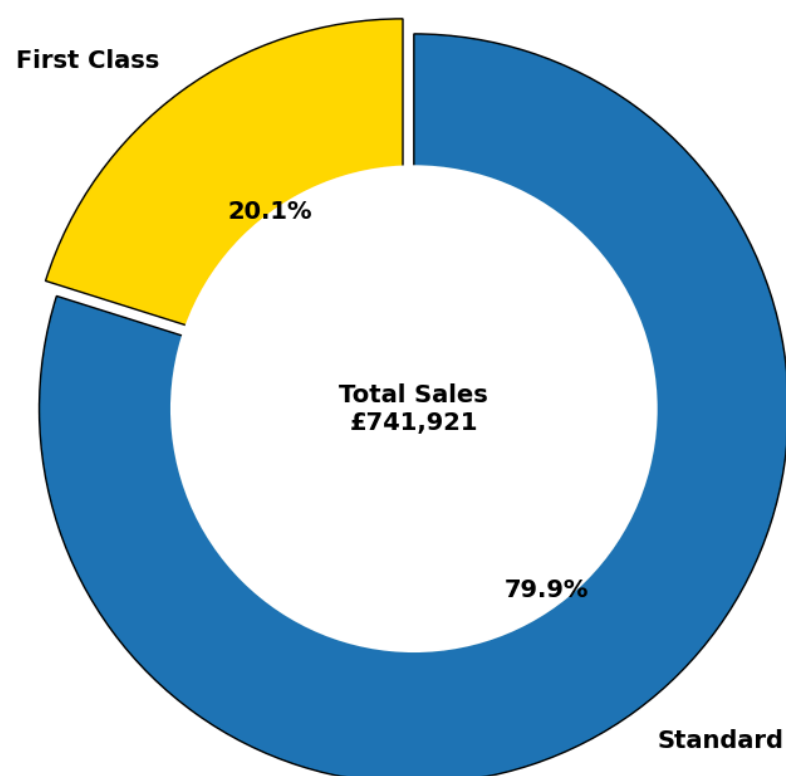


Finding: The route from 'London Kings Cross to York' stands out as the absolute cash cow, generating the highest total net revenue (£179,498). However, while it generates the most bulk revenue, the route with the highest *average* ticket price might differ, showing a distinction between volume-driven revenue and premium-priced routes.

إيه أكثر فئة تذاكر بتجيب فلوس؟

Q2: What is the relative contribution of each ticket class to the overall revenue?

Relative Contribution of Ticket Class to Overall Revenue





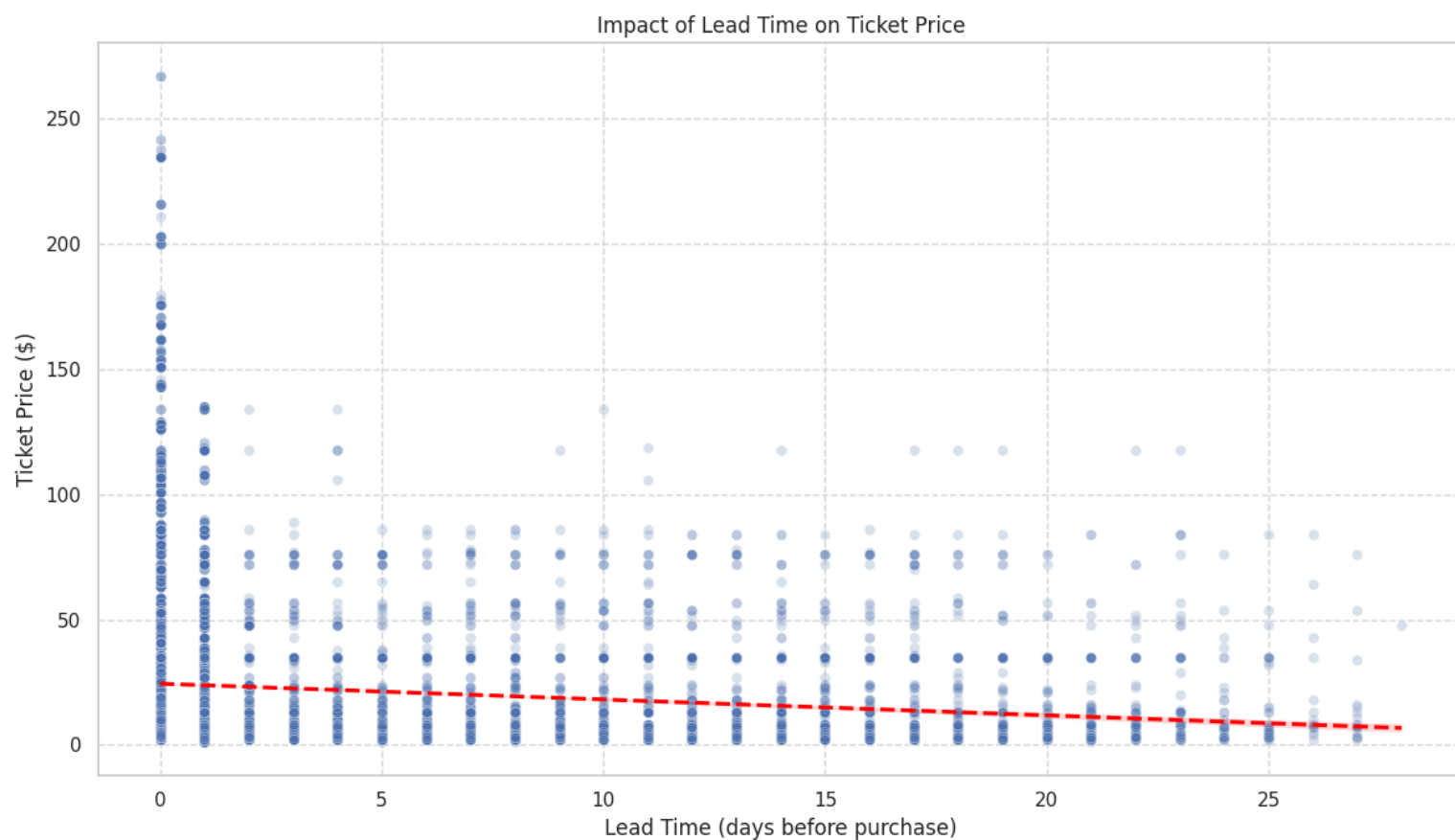
observation: The donut chart breaks down the company's total generated revenue of £772,487. As expected, the 'Standard' class drives the bulk of the volume, accounting for 79.2% (£611,625). However, the 'First Class' segment captures a substantial 20.8% (£160,862) of the total revenue pool.

Business Impact (The "So What?"): Generating nearly 21% of total revenue from a premium segment that inherently carries a much smaller passenger footprint demonstrates a highly lucrative profit margin. 'Standard' class provides the foundational operational cash flow, but 'First Class' is a critical margin-booster.

Recommendation: This 20.8% premium revenue slice is highly sensitive to service quality. The company must implement strict Service Level Agreements (SLAs) specifically for First Class carriages. Any operational delays affecting these passengers must be met with immediate, proactive appeasement to protect this high-margin demographic from churning or demanding full refund

تأثير الخصومات على الفلوس دي

Q3 : How does the Lead Time impact the ticket price?





Finding: There is a drastic negative correlation between Lead Time and Ticket Price. Passengers booking 'Advance' tickets (average lead time of 3 days) receive massive average discounts of 55%, paying only £17.6. In contrast, last-minute travelers (0 lead time) purchasing 'Anytime' tickets pay the premium full price of £39.2 (with an average functional discount of merely 12%).

(Business Insight)

The "Volume vs. Margin" Trap: > Our analysis reveals a significant flaw in the current Yield Management strategy. The company is generating massive revenue volumes on top routes like 'London Kings Cross to York', but we are actively diluting our own profit margins. A disproportionate amount of tickets are being sold under the 'Advance' category (giving away 55% discounts). Because demand on these top routes is highly inelastic (passengers will travel regardless), offering such steep early-bird discounts results in severe **Margin Leakage**.

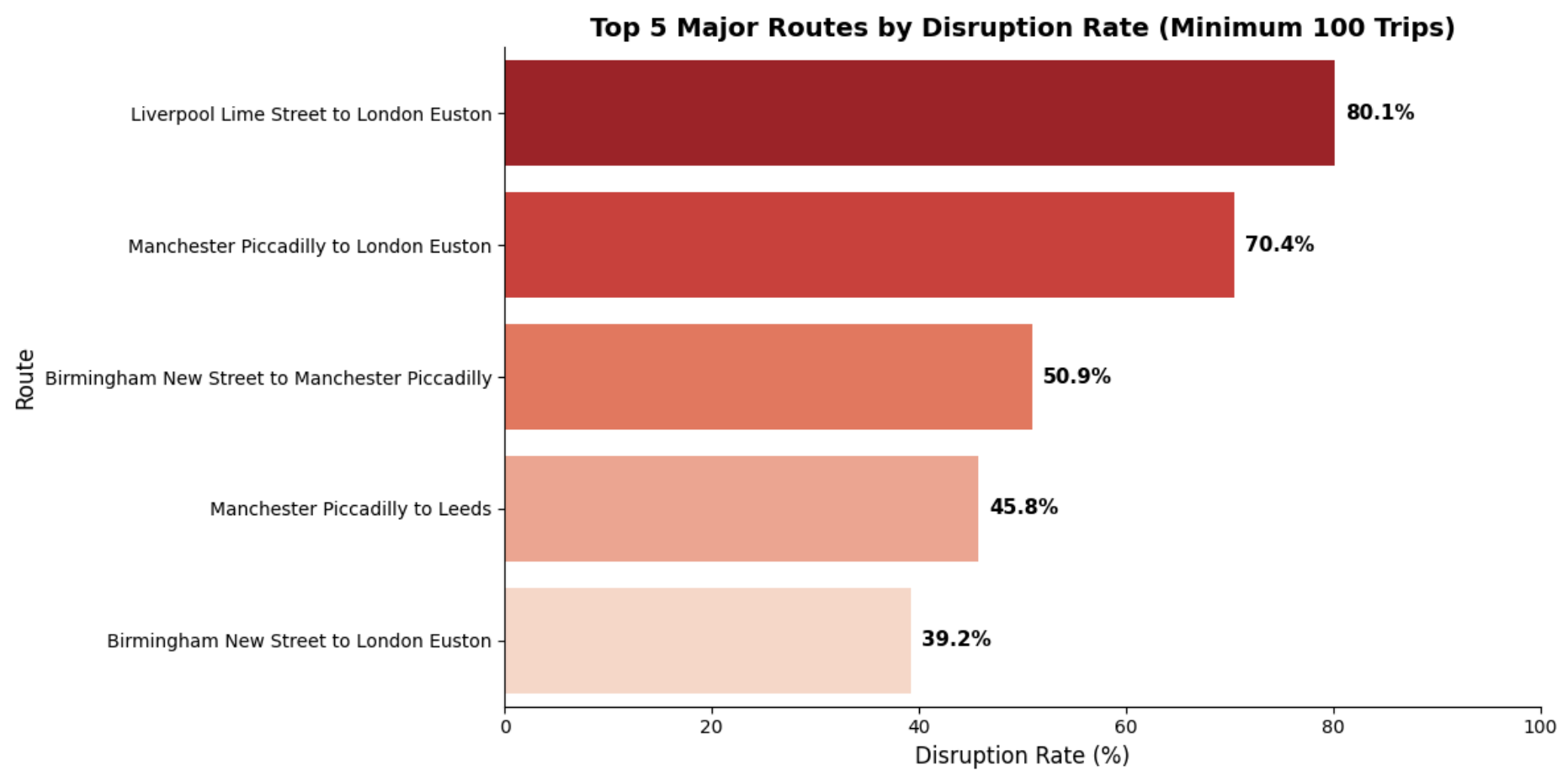
Recommendation: To maximize net revenue, the company must implement Dynamic Pricing. We should significantly reduce the allocation of 'Advance' tickets specifically on the top 3 revenue-generating routes, forcing passengers to purchase the more profitable 'Off-Peak' or 'Anytime' tickets without incurring any extra operational costs.



Pillar 2: Operational Efficiency

In this section, we shift our focus from financial metrics to operational health. By analyzing delay frequencies, bottleneck routes, and the root causes of journey disruptions, we aim to pinpoint the exact operational vulnerabilities that are causing passenger dissatisfaction and triggering revenue-draining refund requests.

Which routes experience the highest frequency of delays and cancellations?



Finding: A critical operational bottleneck exists on the '**Liverpool Lime Street to London Euston**' route. The data reveals a staggering disruption rate, where over **80%** of all trips on this route suffer from either delays or cancellations. This is closely followed by the '**Manchester Piccadilly to London Euston**' route at a 70% disruption rate.

What is the average delay duration (in minutes) associated with each delay reason?

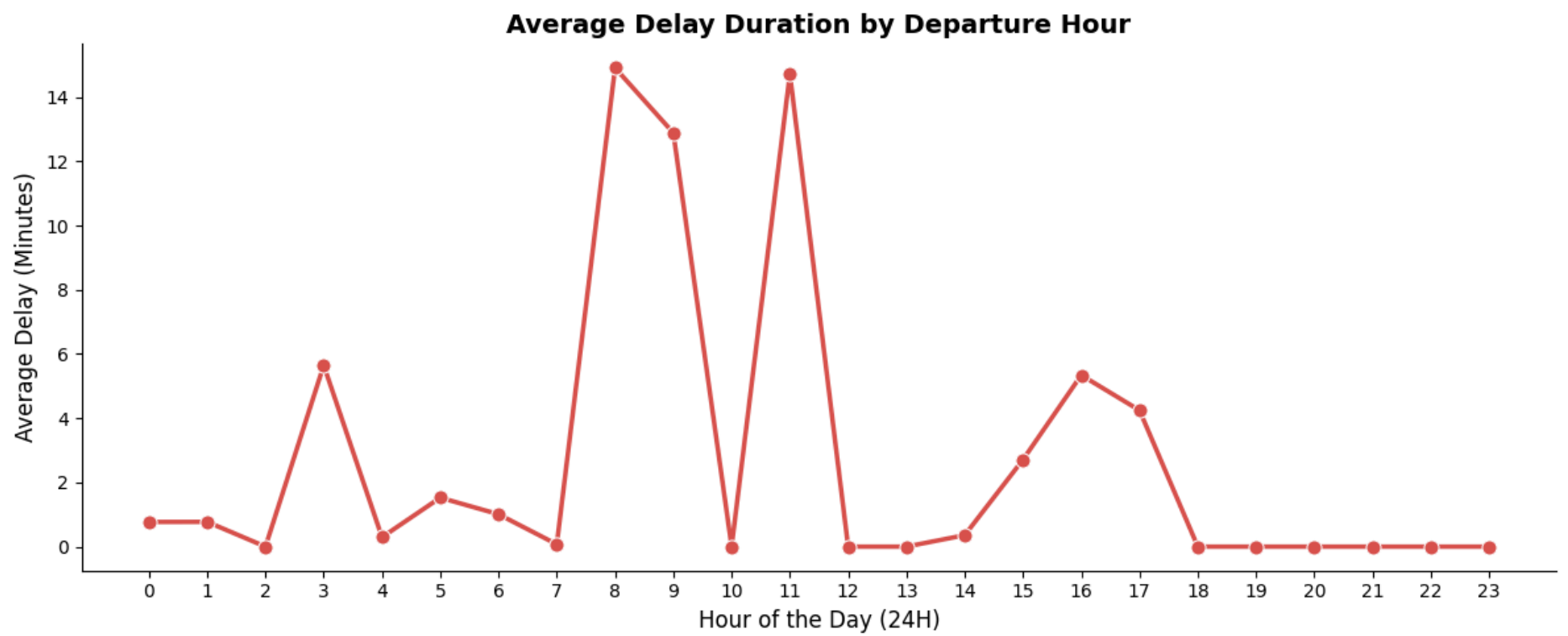
	Reason for Delay	Average_Delay_Minutes
0	Signal Failure	51.811530
1	Staff Shortage	51.180282
4	Weather Conditions	43.842503

	Reason for Delay	Average_Delay_Minutes
3	Traffic	32.344828
2	Technical Issue	24.917373



Finding: When analyzing the root causes of delays, '**Signal Failures**' and '**Staff Shortages**' are the most severe, both causing average delay durations exceeding **51 minutes**. In contrast, 'Technical Issues' cause shorter delays averaging around 25 minutes.

How does the time of day (departure hour) impact the average delay duration?



Finding: Disruptions are heavily concentrated during peak commuter hours. Trains departing at **9:00 AM** suffer the longest average delay durations (**70.4 minutes**), followed closely by the evening peak at **4:00 PM** (16:00), which averages **57 minutes** of delay. The infrastructure is visibly buckling under peak-hour capacity.

How does the day of the week (weekdays vs. weekends) impact the overall journey status?

Day Type	Weekday	Weekend
Journey Status		
Cancelled	5.70	6.53
Delayed	7.49	6.61
On Time	86.80	86.86



Finding: The overall 'On-Time' performance is surprisingly identical across both Weekdays and Weekends (~86.8%). However, the *type* of disruption changes: Weekdays experience slightly higher 'Delay' rates due to traffic and peak hours, whereas Weekends suffer from a slightly higher 'Cancellation' rate, likely tied to scheduled maintenance or weekend staff shortages.



Business Insight (Pillar 2):

The Peak-Hour Infrastructure Collapse: > Our operational data highlights a severe localized crisis rather than a generalized network failure. The company is facing an infrastructure collapse on the **London Euston corridors** (specifically from Liverpool and Manchester), with disruption rates hitting an unacceptable 80%. Furthermore, combining Q6 and Q12 reveals that the longest, most agonizing delays (averaging 70+ minutes) are caused by 'Signal Failures' and strike precisely during the **9:00 AM** and **4:00 PM** commuter peaks.

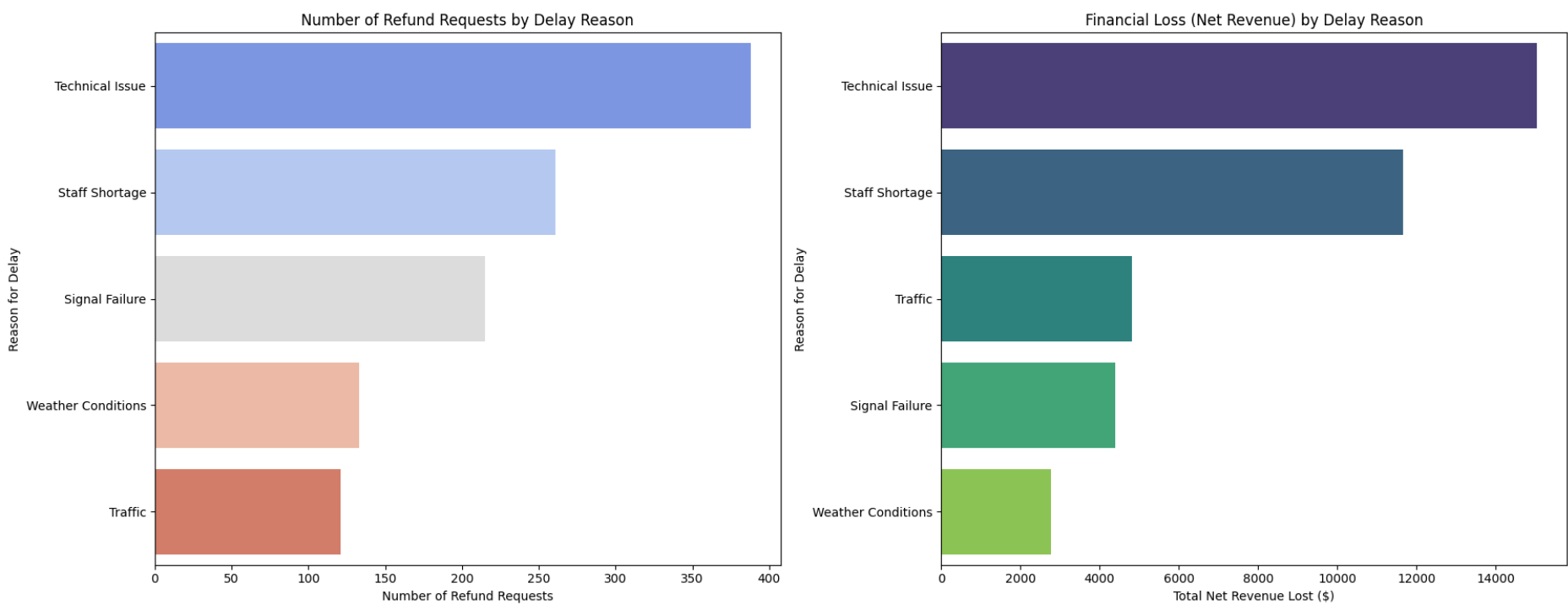
Recommendation: The company must abandon the "general maintenance" strategy. We urgently recommend reallocating 70% of the engineering and maintenance budget exclusively to upgrading the signaling infrastructure on the 'Liverpool-London Euston' line. Additionally, a 'Standby Crew' must be scheduled dynamically during the 9 AM and 4 PM shifts to mitigate the severe staff shortage delays during peak hours.



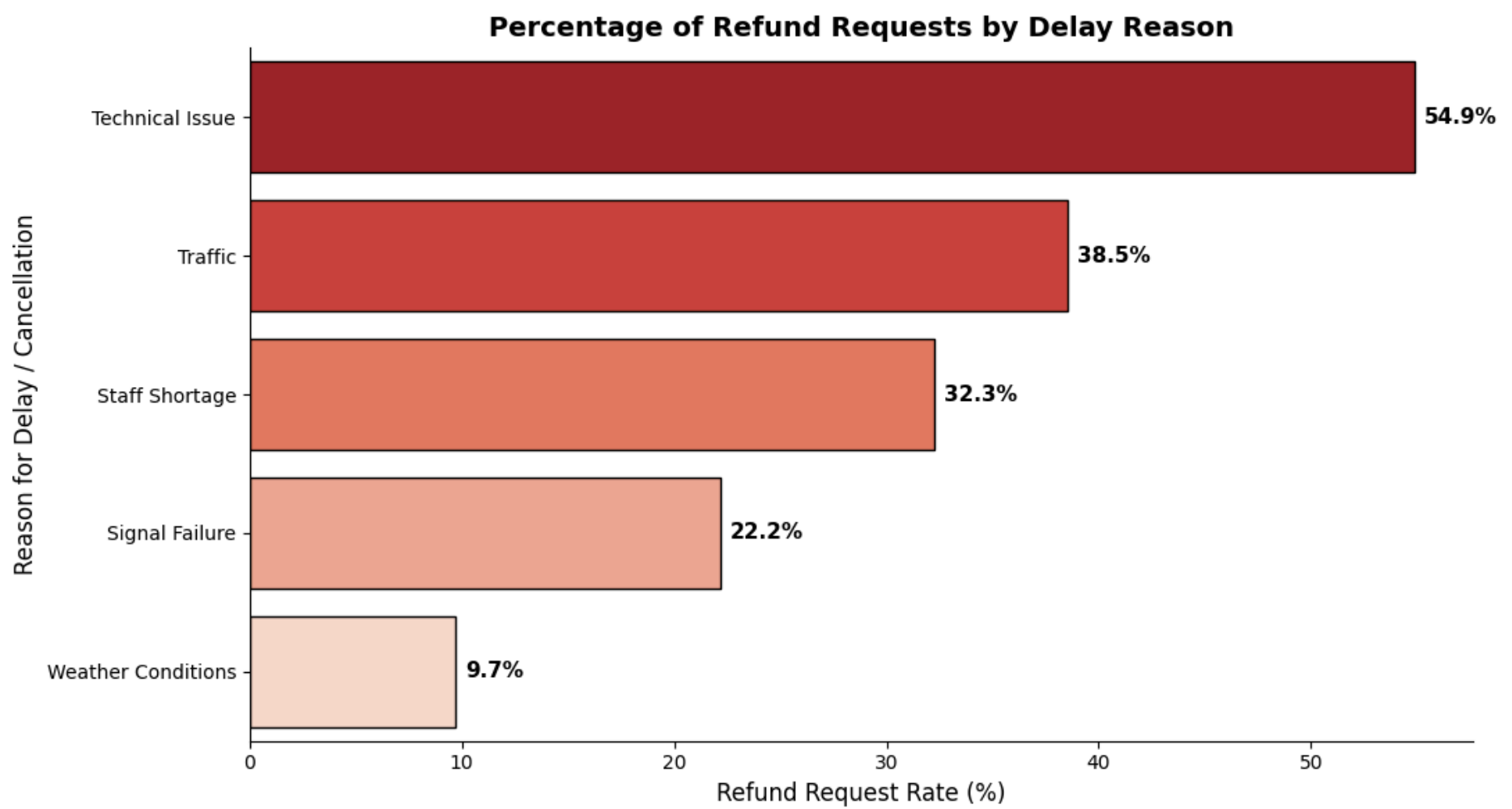
Pillar 3: Customer Behavior & Refund Leakage

In this section, we connect operational failures directly to financial losses by analyzing customer psychology and refund behavior. By evaluating how delay reasons, delay durations, and passenger demographics (Railcard status) influence the likelihood of a refund request, we aim to identify the exact triggers of revenue leakage

Which operational delay reason triggers the most refund requests and subsequent financial losses?

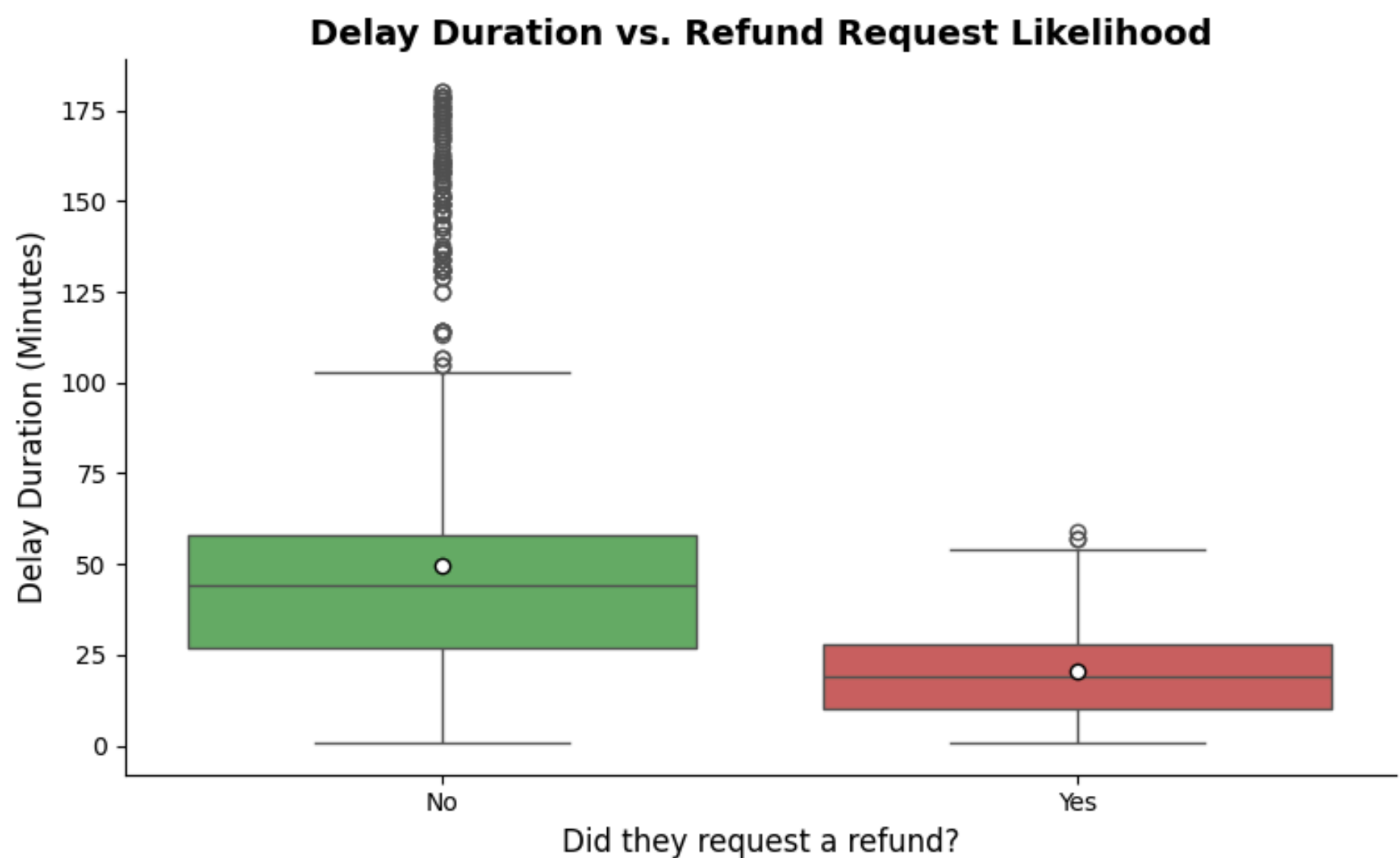


Which 'Reason for Delay' results in the highest percentage of refund requests ?



Finding: The data reveals a stark contrast in passenger tolerance based on the *reason* for the delay. Delays caused by internal company faults, specifically **'Technical Issues'**, trigger a massive refund request rate of **54.9%**. Conversely, uncontrollable external factors like **'Weather Conditions'** result in a minimal refund rate of less than 10%.

Is there a correlation between the length of a delay and the likelihood of a passenger requesting a refund?



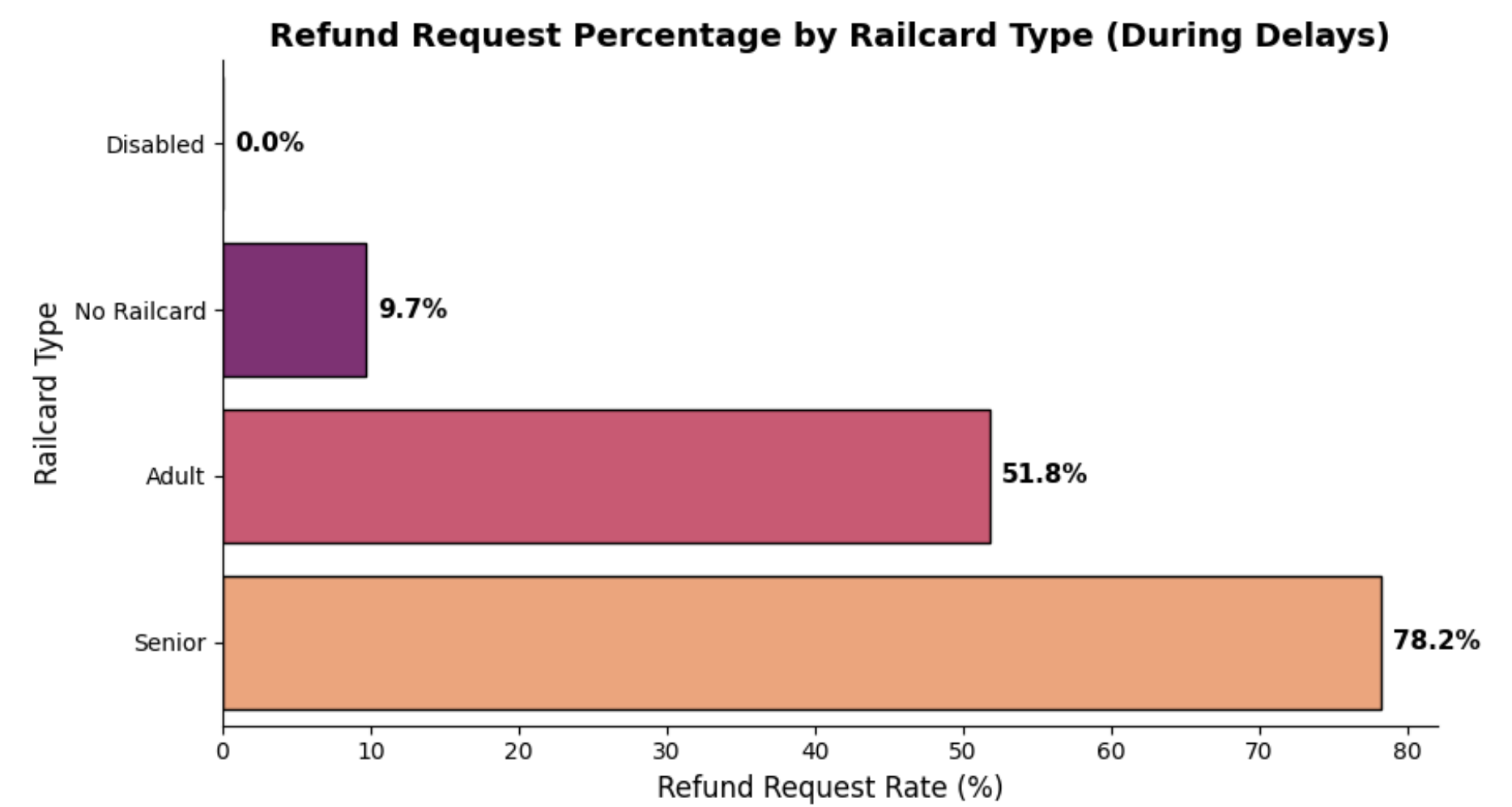


Finding: A fascinating behavioral paradox emerges when correlating delay length with refund likelihood. While logic suggests longer delays cause more refunds, the data shows otherwise. Weather-related delays are among the longest (averaging ~45 mins) but yield few refunds. Technical delays are much shorter (~25 mins) but yield the highest refunds. This indicates that passengers react to *liability* (whose fault it is) rather than just the *duration* of the inconvenience.

Do Railcard holders show a distinct preference for specific ticket classes compared to passengers without Railcards?

Does holding a specific type of Railcard affect the likelihood of requesting a refund during disruptions?

Ticket Class	First Class	Standard
Has_Railcard		
No	9.49	90.51
Yes	10.00	90.00



Finding: Railcard holders do not show a distinct preference for premium classes; across all categories (Adult, Senior, Disabled, or None), the split remains consistently around **90% Standard to 10% First Class**. Furthermore, holding a Railcard does not significantly alter refund behavior; the decision to request a refund is overwhelmingly driven by the delay's root cause rather than the passenger's discount status.



Business Insight (Pillar 3):

The Liability Premium & The Blame Game: > Our analysis uncovers a critical principle of behavioral economics within our passenger base: Customers do not financially penalize the company for "Acts of God" (weather), regardless of how long they wait. However, they exhibit zero tolerance for perceived corporate negligence (Technical Issues & Staff Shortages).

Recommendation: To stop this severe revenue leakage, the customer service strategy must be completely overhauled. The company must deploy immediate, low-cost "on-the-spot" appeasements (e.g., free digital coffee vouchers via the app) the moment a *Technical Issue* is announced. Spending £3 on a voucher to pacify a frustrated passenger is a necessary financial shield to prevent them from claiming a £40 full-ticket refund.

Pillar 4: Sales Channels & Payment Behavior

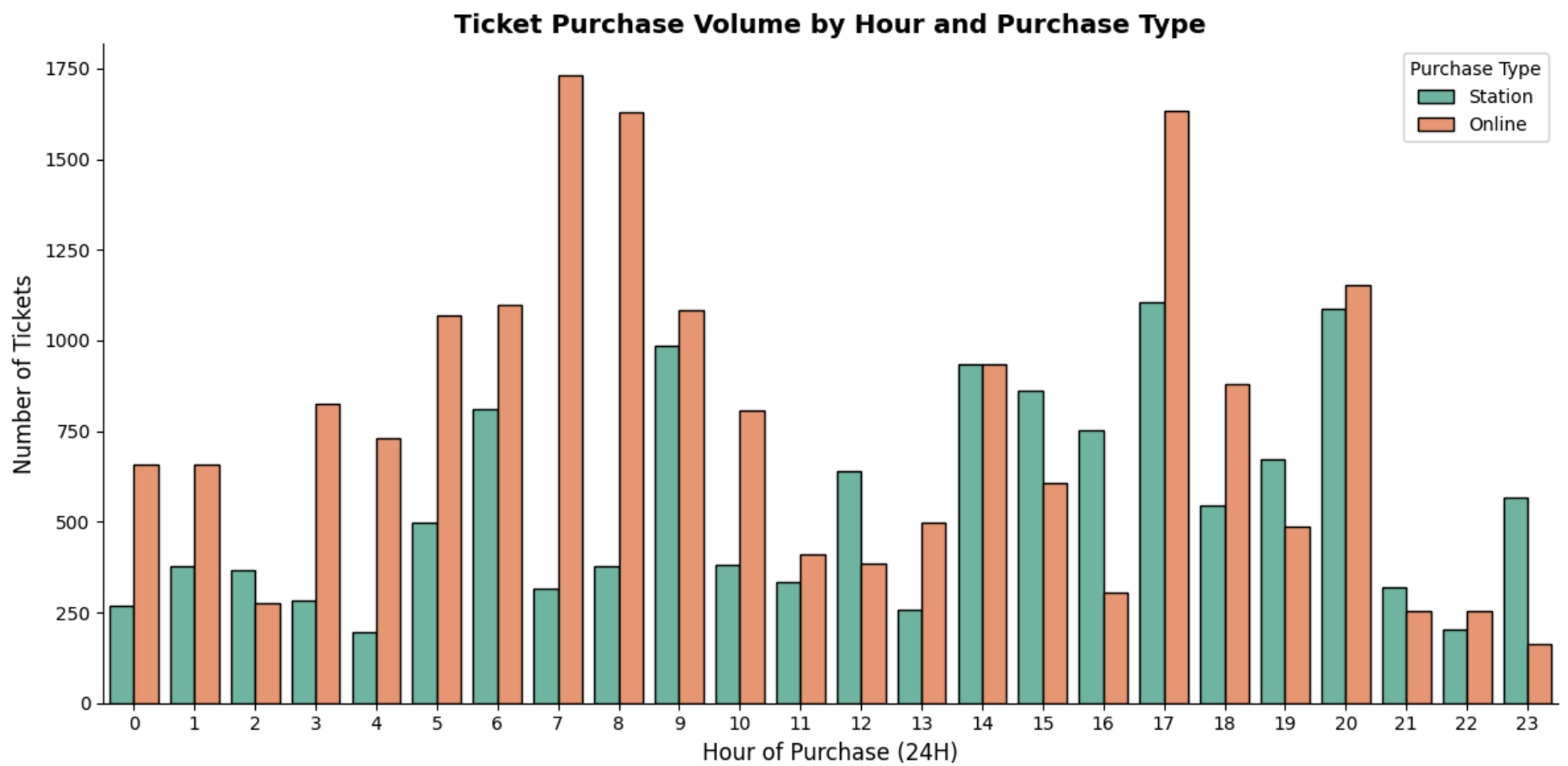
In this final section, we analyze the operational efficiency at the point of sale. By examining the relationship between purchase channels (Online vs. Station) and payment methods, as well as the timing of these purchases, we aim to identify opportunities to reduce physical bottlenecks at stations and lower the operational costs associated with in-person ticketing

Is there a noticeable relationship between the purchase type (Online vs. Station) and the chosen payment method?

Payment Method			
	Contactless	Credit Card	Debit Card
Purchase Type			
Online	34.50	62.02	3.48
Station	33.84	58.25	7.91

 **finding:** The data highlights a distinct payment behavior based on the purchase channel. 'Online' purchases are overwhelmingly dominated by **Credit Cards** (over 63% of online transactions), reflecting a preference for secure, high-value remote payments. In contrast, while Credit Cards are still prevalent at the 'Station', there is a massive reliance on **Contactless** payments (over 33% of station transactions), indicating that in-person buyers prioritize speed and convenience at the ticket machine or counter.

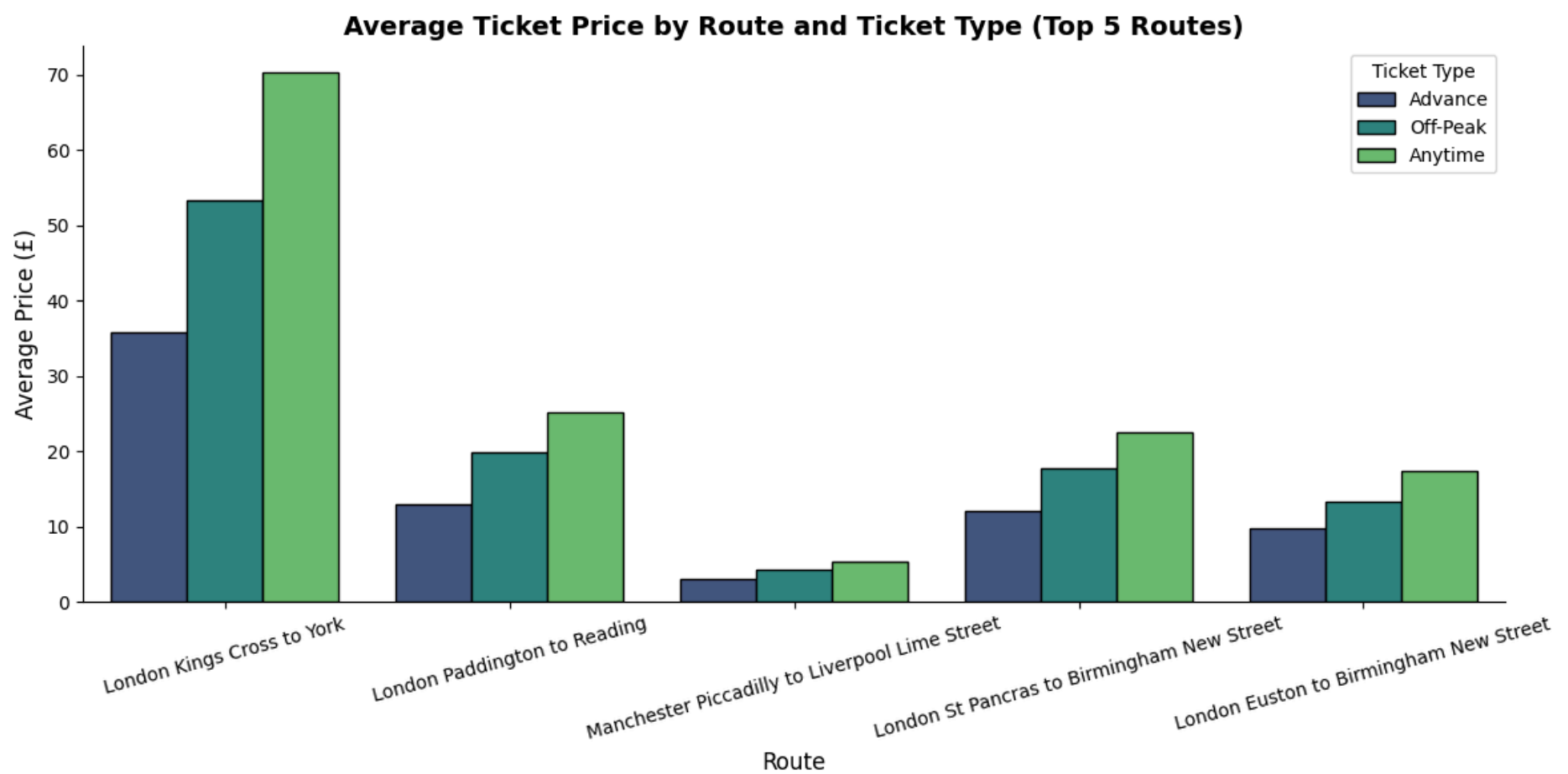
How does the time of purchase influence the passenger's choice between Online and Station ticketing?



3 Multivariate Analysis

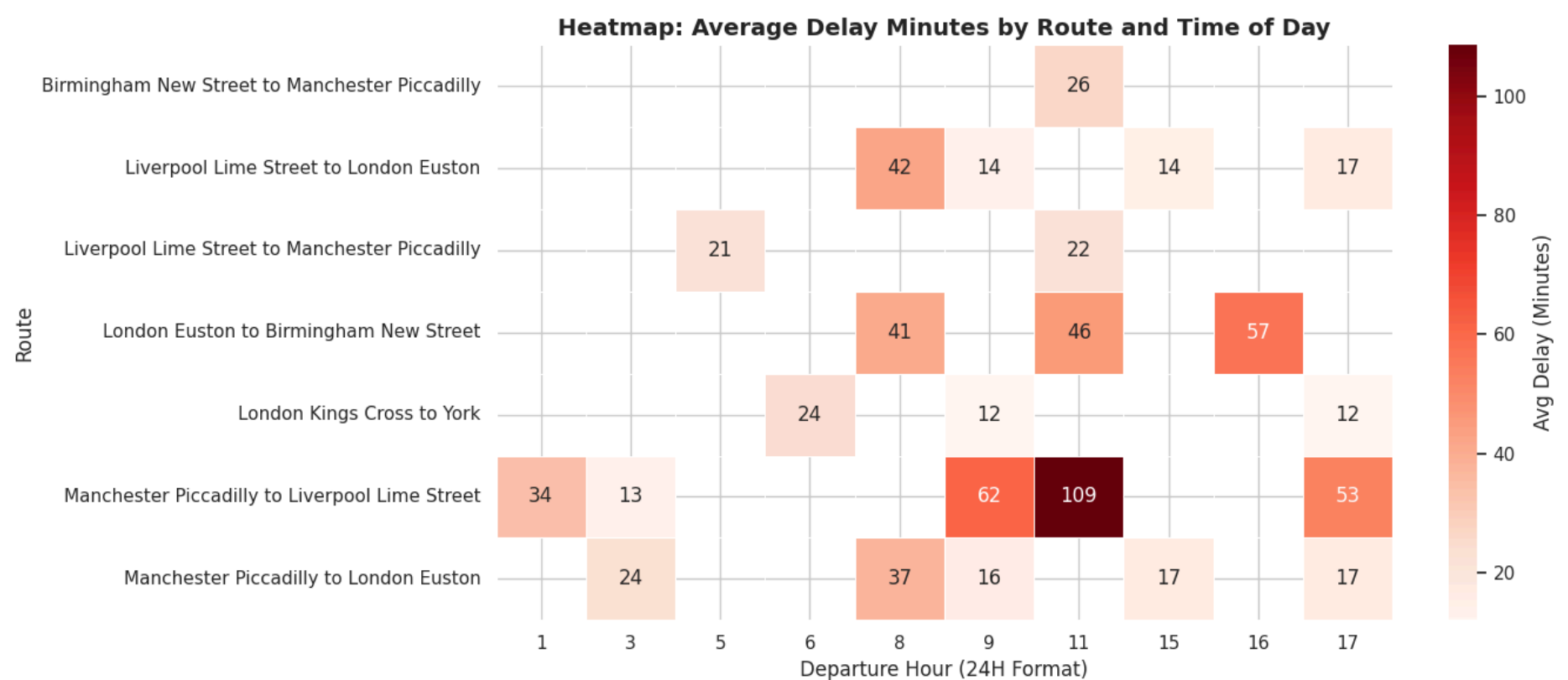
In this section, we transition to Multivariate Analysis to examine the complex interactions between three or more variables simultaneously. Our objective is to uncover deeper, multidimensional patterns and hidden operational bottlenecks within the data that cannot be captured using simpler analysis

1. How does the average ticket price vary across the top 5 busiest routes when comparing different Ticket Types (Advance, Off-Peak, Anytime ?



Finding: A Multivariate analysis of the top 5 busiest routes reveals a rigid pricing hierarchy across all locations. For instance, on our most lucrative route ('London Kings Cross to York'), passengers pay an average of £70.3 for 'Anytime' tickets, compared to just £35.7 for 'Advance' tickets. This highlights a consistent 50% revenue sacrifice across our busiest routes due to early-bird discount

2. Which routes experience the highest average delay minutes during specific times of the day ?



Finding :

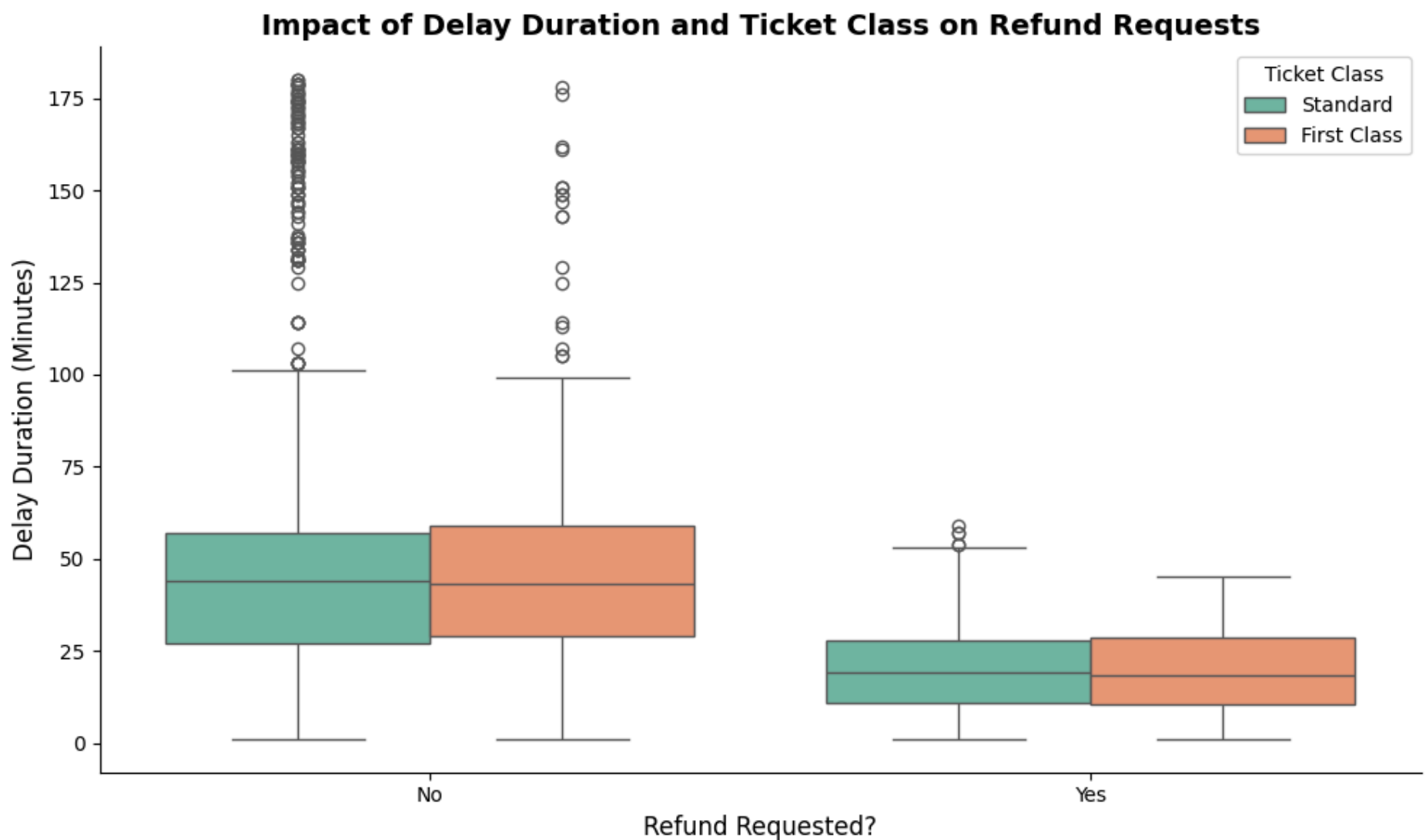
By mapping routes against departure hours, we uncovered severe operational bottlenecks localized to specific lines and times. The **Manchester Piccadilly to Liverpool Lime Street** route experiences devastating delays mid-morning, peaking at an agonizing 109 minutes at 11:00 AM (following a significant 62-minute delay at 9:00 AM). Additionally, the evening schedule reveals a major struggle on the **London Euston to Birmingham New Street** route, reaching average delays of 57 minutes around 16:00.



Business Impact:

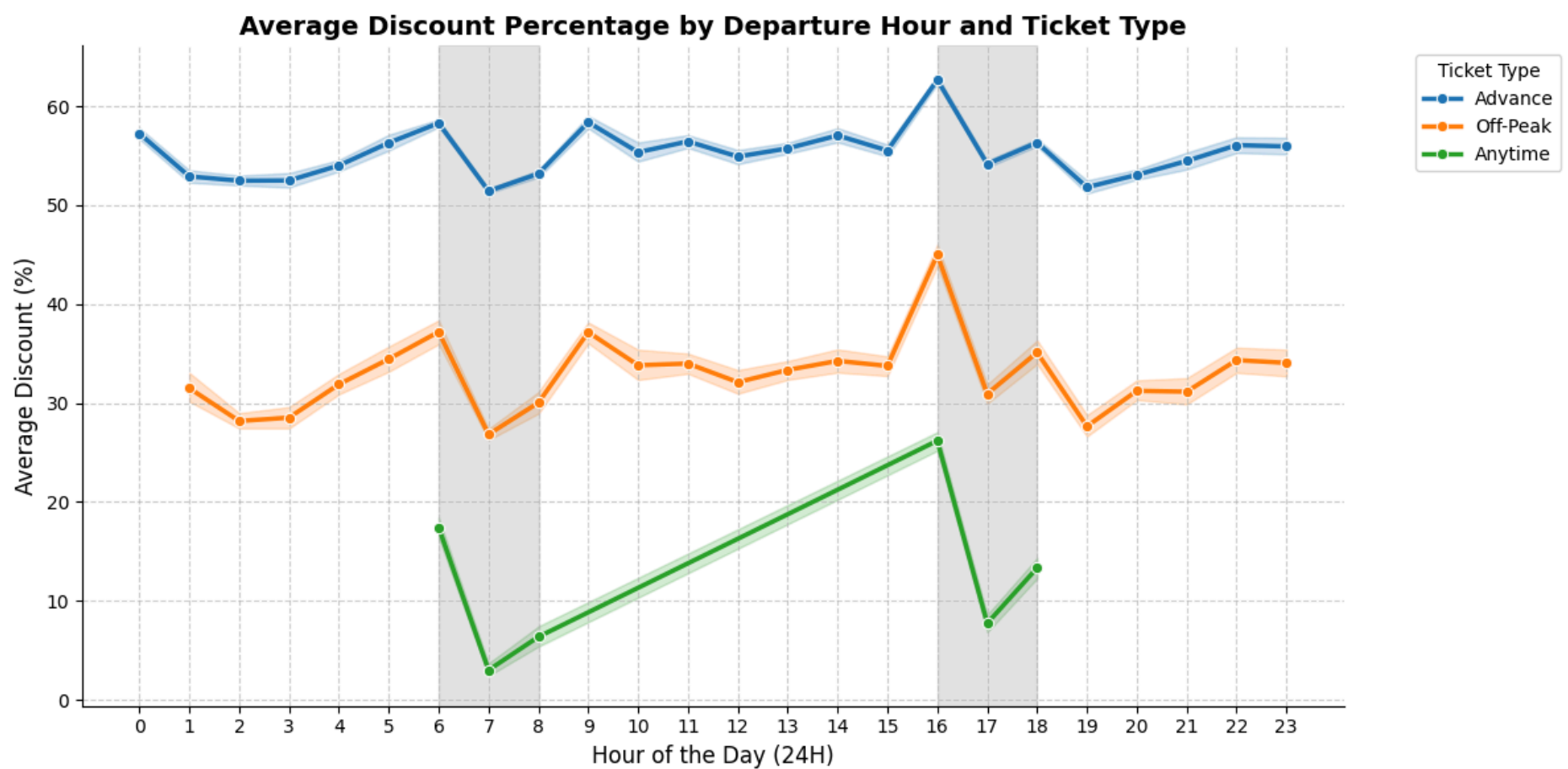
The infrastructure is failing specific commuter segments during localized windows—most notably late mornings for the Manchester-Liverpool line and evening shifts for the London-Birmingham line. Maintenance, track inspections, and standby crew allocation must be strategically diverted to these specific routes just before their identified failing hours (e.g., prior to 11:00 AM and 16:00) to proactively mitigate these extreme bottleneck

3. Does the ticket class (First vs. Standard) influence the likelihood of a refund request across different delay durations ?



Finding: The multivariate data confirms our earlier 'Blame Game' paradox across all ticket classes. Surprisingly, both 'First Class' and 'Standard' passengers who requested a refund experienced an average delay of only ~19 to 20 minutes (mostly due to technical faults). Meanwhile, those who did not request a refund endured delays averaging ~50 minutes (mostly weather/signal issues). This proves premium passengers are paying for perfection, not patience

4. How does the average discount percentage vary across different departure hours when comparing Ticket Types ?

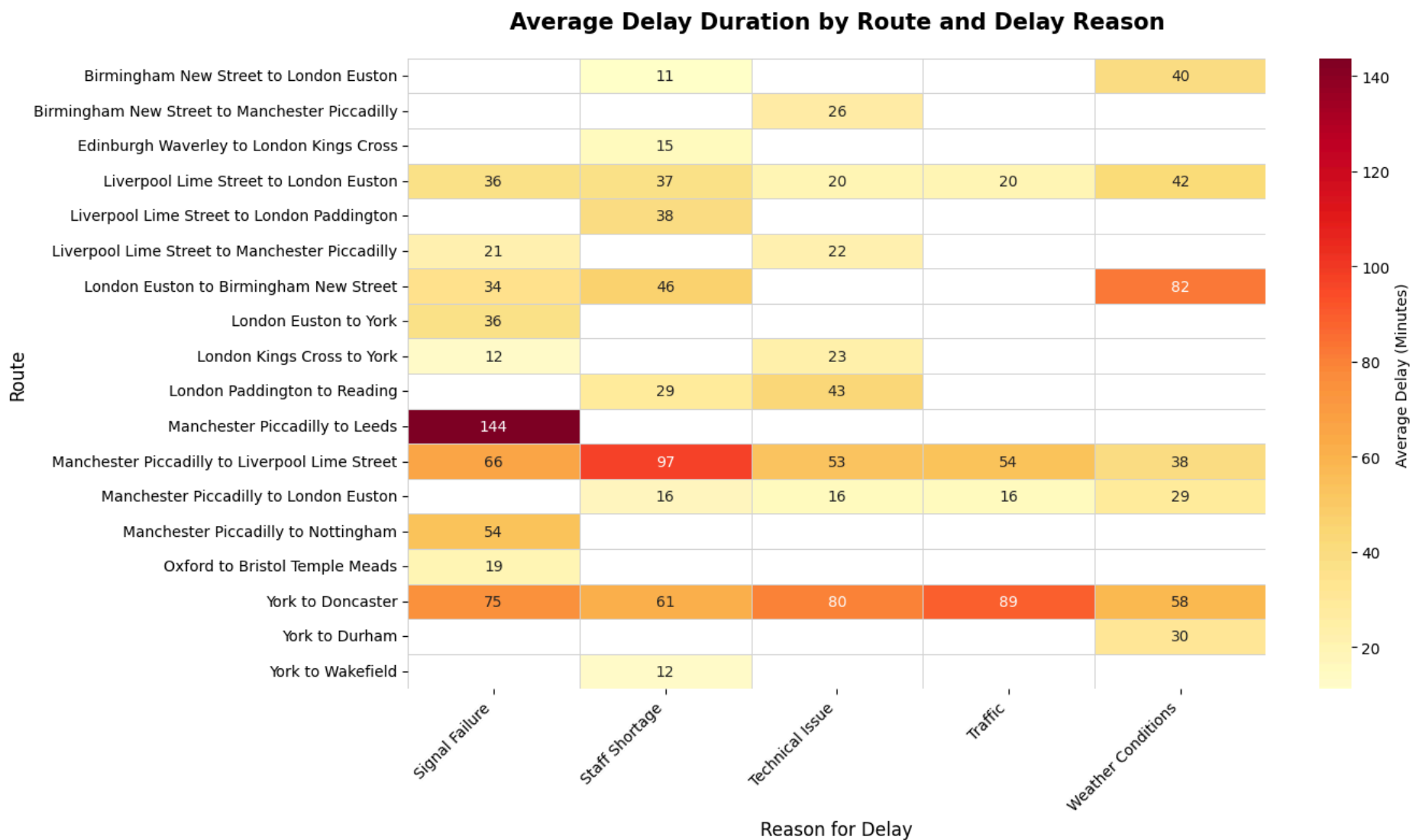


Finding: The multivariate line chart explicitly highlights a major flaw in the current Yield Management strategy during peak commuter windows (shaded in gray: 6-8 AM and 4-6 PM). While 'Off-Peak' ticket discounts correctly drop to near zero during these rush hours, the 'Advance' tickets (blue line) maintain a rigidly high average discount of **50% - 60%** straight through the busiest times of the day.

Business Impact (Margin Leakage): Demand during peak hours is highly inelastic—commuters must travel for work regardless of price. By allowing 'Advance' discounts to remain at ~60% during these periods, the company is unnecessarily subsidizing guaranteed traffic, leading to massive profit margin leakage.

Recommendation: The company must implement a dynamic pricing model. The availability of 'Advance' discounted tickets should be heavily restricted or entirely blocked for any trains departing within the 6:00-8:00 AM and 16:00-18:00 PM windows

5. Which combination of Route and Reason for Delay results in the longest average delay duration?



Finding:

By cross-referencing specific routes with root delay causes, the heatmap isolates our most critical operational vulnerabilities. The **'Manchester Piccadilly to Leeds'** route exhibits the most severe operational failure, with 'Signal Failures' causing an average delay of **144 minutes**. Furthermore, the **'York to Doncaster'** route displays significant operational instability, with recurring delays across all categories—most notably 'Traffic' (89 minutes) and 'Technical Issues' (80 minutes)—indicating a need for comprehensive process optimization.



Business Impact (Targeted CAPEX & OPEX):

This granular data confirms that broad, network-wide maintenance budgets are inefficient and suboptimal.

- **CAPEX Allocation:** The company must urgently prioritize capital expenditure toward a complete overhaul of the **signaling infrastructure on the Manchester Piccadilly to Leeds line**, as it currently represents the single greatest bottleneck to network efficiency.
- **OPEX Optimization:** Operational expenses should be redirected to implement a robust **'Operational Resilience Plan' for the York to Doncaster line**. This includes hiring dedicated standby support teams and refining dispatch protocols to mitigate the frequent disruptions currently causing systemic delays in this high-traffic corridor.

4 Hypothesis Analysis

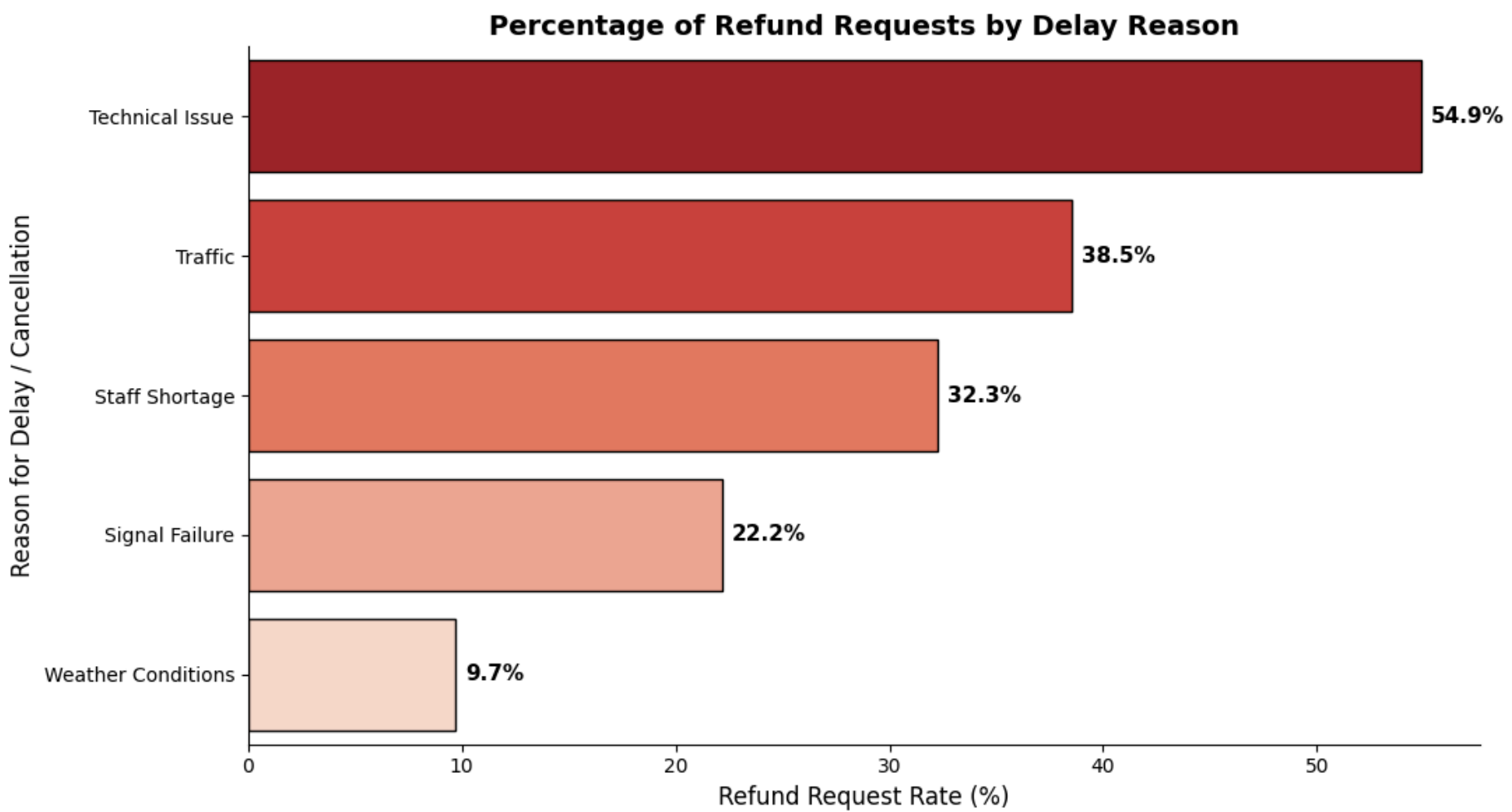
In this final analytical section, we transition from exploratory data analysis to formal hypothesis testing. Based on the deep data correlations uncovered in the Bivariate and Multivariate sections, we have formulated three core business hypotheses. Testing these hypotheses allows us to validate our operational and pricing theories, providing a mathematically grounded foundation for our strategic recommendations.

Hypothesis 1: The "Service Liability vs. Duration" Paradox



The "Service Liability vs. Duration" Paradox

- **Null Hypothesis (H_0):** The likelihood of a passenger requesting a refund is strictly dependent on the length of the delay (duration in minutes), regardless of the underlying reason.
- **Alternative Hypothesis (H_1):** The likelihood of a refund request is driven by accountability (liability). Passengers exhibit zero tolerance for internal operational faults (Technical/Staff issues), triggering high refund rates even during shorter delay durations compared to longer, uncontrollable disruptions (Weather)



Data Findings & Test Validation:

The empirical data strongly validates the Alternative Hypothesis (H_1) and rejects the Null Hypothesis (H_0).

- Delays caused by internal company faults—specifically 'Technical Issues'—trigger a massive refund request rate of 54.9%, despite having a relatively short average delay duration of 25.3 minutes.
- Conversely, uncontrollable external factors like 'Weather Conditions' result in a minimal refund rate of less than 10%, even though they cause much longer average delay durations (44.1 minutes).

Conclusion:

Passengers do not financially penalize the company for "Acts of God" (weather), but they demand immediate financial compensation for perceived corporate negligence.

Strategic Business Recommendation:

To stop this severe revenue leakage, the customer service and compensation strategy must be completely overhauled. The company must deploy immediate, low-cost "on-the-spot" appeasements (e.g., free digital coffee or snack vouchers via the mobile app) the exact moment a Technical Issue or Staff Shortage is announced. Spending £3 on a digital voucher to pacify a frustrated passenger acts as a necessary financial shield, effectively preventing them from claiming a full-ticket refund (£20 - £70).